A project proposal for partial fulfilment of the course unit IT3162 Group Project for the degree of Information Technology

SmartLab Guardian

(Smart Computer Laboratory Management & Monitoring System)

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Declaration

We hereby declare that the project proposal submitted for evaluation of the course module IT3162 leading to the award of a Bachelor of Science in Information Technology is entirely our own work, and the contents taken from the work of others have been cited and acknowledged within the text. This proposal has not been submitted for any degree at this University or any other institution.

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Recommendation of the Supervisor

I recommend that the project be carried out by the students under my supervision,

……………………………. ……………………………..

Ms. S.Bramyah Date
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1.Introduction

1.1 Introduction

SmartLab Guardian is a centralized Computer Laboratory Management and Monitoring System designed to improve laboratory administration by detecting possible student account misuse, simplifying examination computer assignments, managing hardware inventory, providing alerts for disconnected devices, and monitoring computer usage. By integrating these functions into one system, the project aims to reduce manual work, save time, and improve the efficiency and security of university computer laboratories.

1.2 Aim and Objectives

Aim

To design and develop a centralized computer laboratory management and monitoring system that improves security, automates routine administrative tasks, and gives laboratory administrators real-time visibility into the state of the laboratory.

Objectives

- Detects student login and identity misuse through computer activity.
- Automate examination computer assignments using Excel data.
- Maintain centralized laboratory hardware inventory records.
- Detect and alert unauthorized hardware disconnections.
- Monitor computer sessions, applications, and website activity.
- Generate reports on usage, attendance, and hardware status.

1.3 Benefits of this system

- Reduces laboratory staff workload through automation.
- Improves examination security by detecting account misuse.
- Reduces examination preparation time.
- Provides alerts for missing or disconnected hardware.
- Monitors student computer activity and engagement.
- Maintains accurate and secure attendance records.
- Supports better laboratory management through consolidated reports.

2.Background

2.1 Background

Currently, the university uses Classroom Spy to monitor student computers, while several laboratory management tasks are handled manually or through separate systems. This creates difficulties in detecting account misuse, assigning students to computers during examinations, managing hardware inventory, detecting disconnected devices, and reviewing computer usage. SmartLab Guardian is proposed as a centralized system to automate these tasks, monitor student and computer activity, manage hardware, generate alerts, and produce useful reports, thereby reducing manual work and improving laboratory management efficiency.

2.2 Review of Existing Systems

Computer laboratories currently use a combination of monitoring software, manual procedures, spreadsheets, and separate records. These approaches have several limitations:

- **Monitoring Software:** Tools like Classroom Spy monitor student computers but do not cover complete laboratory management.
- **Manual Examination Setup:** Student-to-computer assignment and login preparation are time-consuming.
- **Separate Hardware Records:** Spreadsheets make hardware tracking and management difficult.
- **Manual Hardware Checks:** Disconnected or removed devices may not be detected immediately.

3.Materials and Methods

3.1 Proposed system design

The proposed system, SmartLab Guardian, is designed as a centralized computer laboratory management and monitoring system.It integrates student login monitoring, examination computer assignment, hardware management, hardware removal detection, and computer usage monitoring.

User Roles:

- Admin : Manages students database,manages hardware database and responds to hardware removal alerts.
- Lecturer :Start session,end session,view student attendance
- Examiner :Signs in students to assigned pc, and views relevant examination reports.
- Student: :Log in to pc with username and password

Student Login & Identity Misuse Detection:

- Records student login and computer details.
- Monitors application, website, and active computer usage.
- Generates alerts for predefined suspicious login or activity conditions

Remotely Manage Student Examination Login:

- Imports student and computer details through Excel.
- Automatically assigns students to available computers.
- Reduces manual examination setup and login time.

Hardware Inventory Management:

- Maintains records of keyboards, mice, monitors, system units, and other equipment.
- Stores hardware ID, PC number, condition, and status.
- Tracks available, assigned, damaged, and missing equipment.

Hardware Removal Alert:

- Monitors registered hardware connected to each PC.
- Detects unexpected device disconnection.
- Generates alerts containing PC number, device, date, and removal time.
- Maintains hardware removal history.

Computer Usage & Session Monitoring:

- Records login, logout, active usage time, application usage, and website activity.
- Helps identify inactive, excessive, or unauthorized computer usage.
- Generates time-based usage reports.

System Components:

- Admin Dashboard: Management, monitoring, alerts, and reports.
- Student PC Monitoring: Monitoring Agent Collects computer usage, login, and hardware information.
- Backend Server: Processes events and applies predefined monitoring rules.
- Database: Stores students, computers, hardware, sessions, alerts, and reports.

3.2 Functional Requirements

Automated Attendance Marking

- Student login to the system will be recorded as attendance. The system will continuously monitor the student's computer activity and evaluate it against predefined requirements, such as minimum active time and the use of required applications. Students whose activity does not meet the specified criteria will be flagged for further review.

Fast Automatic Sign-In For Examination

- The system automatically signs in the relevant student to their assigned PC.
- This removes the need for lab staff to walk to each PC and log students in individually, significantly reducing setup time.

Hardware Inventory Management

- The system checks connected hardware (mouse, keyboard, hard drive, etc.) on each PC every 5 minutes.
- It compares the current check against the previous report to detect any change, removal, or disconnection of hardware.
- If a change is detected, the system raises a notification to the admin and logs it in a hardware status report.

Computer Usage & Application Monitoring

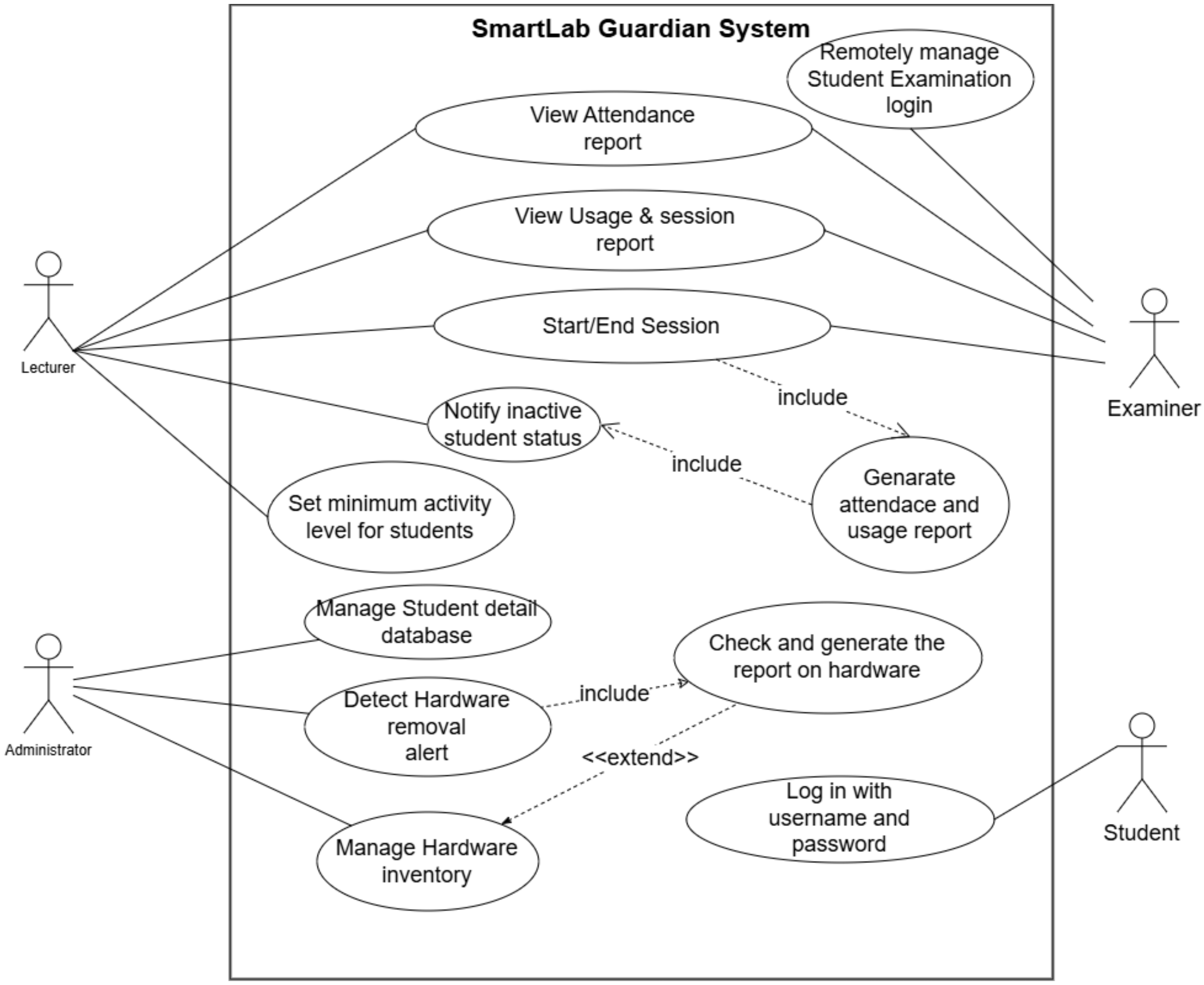
- The system tracks each student's application usage and screen time during the session.
- A complete usage report per student (apps used and time spent) is generated after the session.

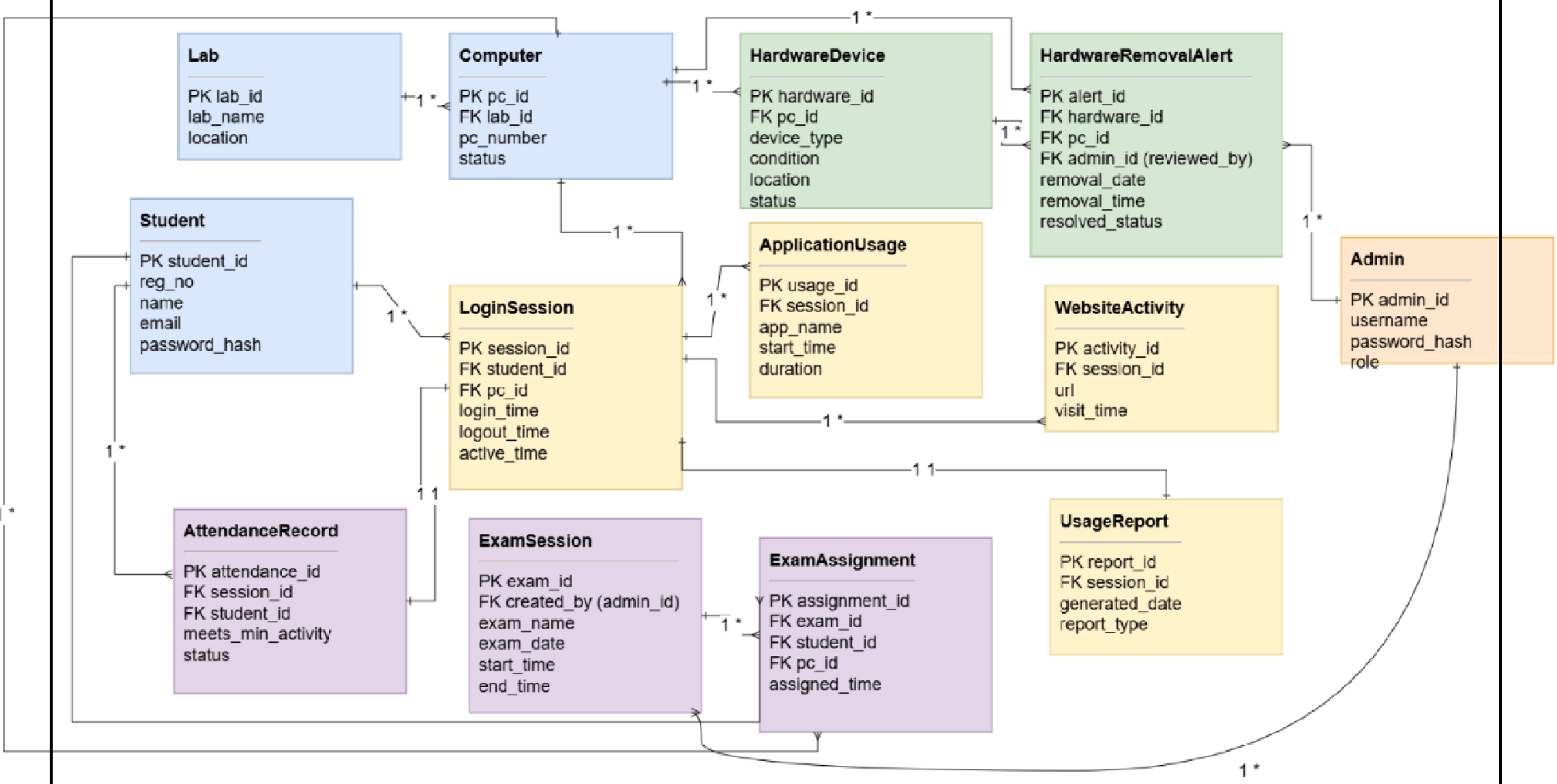
3.3 Non-functional Requirements

- **Performance:** The system will record attendance upon student login and evaluate their activity based on predefined time and application usage requirements. Students with insufficient activity will be flagged for review.
- **Reliability:** The monitoring system must continue collecting attendance, login, and usage data through brief network interruptions and resynchronize with the server once connectivity is restored, so no session data is lost.
- **Scalability:** The system should support all computers in a lab, and multiple labs at once, without a noticeable drop in dashboard responsiveness as more PCs and students are added.
- **Security:** Login credentials, and session/usage data must be stored and transmitted securely to prevent tampering or unauthorized access.
- **Usability:** The admin dashboard should present attendance, sign-in status, hardware alerts, and usage reports in a clear, simple format requiring minimal training.
- **Availability:** Attendance marking, auto sign-in, hardware checks, and usage monitoring should remain available throughout scheduled lab hours, including exam periods.

3.4 Use case diagram & ER diagram

The diagram below summarizes the main actors – Student, Examiner, and Lab Administrator and the primary use cases they interact with in SmartLab Guardian.

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removal_time
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assigned_time
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3.5 Tools and Technologies

The following technology stack is proposed for the implementation; specific choices may be refined during development.

- **Backend :**

  Node.js - Server-side runtime for handling requests and business logic.

  Express.js - Web framework for building RESTful APIs and managing routing.

- **Frontend :**

  React.js- For building dynamic and responsive UI.

  HTML & CSS & JavaScript - core web technologies for structure and interactivity.

- **Database:**

  Mongo DB Atlas - Cloud platform that allows to deploy , manage, secure and scale mongo DB databases.

- **Version Control & Collaboration:** Git and GitHub

- **Other Tools:**

  Postman - For testing and verifying backend APIs.

  VS Code - Integrated development environment for coding and debugging.

4. Expected Results

4.1 Brief description of the expected results

- **Monitoring System:** Installed on laboratory computers to monitor user sessions, system status, and activities.
- **Suspicious Login Detection:** Identifies accounts that are logged in but show activity inconsistent with genuine student use.
- **Automated Exam Setup:** Assigns computers, seats, and student logins automatically, reducing exam preparation time to a few minutes.
- **Hardware Inventory:** Maintains an updated list of laboratory computers and hardware.
- **Device Alerts:** Automatically alerts administrators when a registered computer/device becomes disconnected.

- **Centralized Dashboard:** Provides administrators with a single interface for monitoring and management.
- **Reports & Analytics:** Generates **attendance, computer usage, login, and activity reports** for daily supervision and periodic review.
- **Prototype Demonstration:** Tested on a representative group of laboratory computers, with the system designed to scale to the **entire laboratory**.

5. Timeline of the Project

The project is planned over a 12-week schedule, from initial requirements gathering through to final documentation and submission, as shown below.

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Requirements & literature review												
System & database design												
Use case & UI/UX design												
Monitoring system development												
Dashboard & backend API												
Hardware alert & attendance integration												
System integration & testing												
Documentation & report writing												
Final review & submission												

6. GitHub repository information

The source code, documentation, and version history for this project are maintained in a public GitHub repository:

Repository:
https://github.com/RaneemaRisnee/SmartLab-Guardian

7. References

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3.Build a Simple File Transfer System Using Python (Network Programming Project) | by Nathan Chiche | Medium

4.How to Choose a Tech Stack in the AI Era - Full Scale